

Engineering What an Optical Sensor Cannot See

[AUTHORS]

[AFFILIATION]

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Abstract. A symmetry of an optical measurement law makes a scene subspace exactly invisible under an idealized band-limited model; on an unfiltered bench that wall becomes a calibrated $\sim 7 \times 10^{-3}$ residual, suppressed by near-conjugate relay imaging (Section 6). We turn that blindness into a calibrated, certifiable design variable. A physics mapping assigns each optical symmetry class its wall type—a support wall and a lossless-bucket energy wall, exact to machine precision in the model (leaks near 10^{-16})—with the one controlled breaker that reopens it. A measured certification apparatus turns “cannot see” into a singular spectrum $\beta_d = \sigma_d$: 80 code dimensions are certified blind at a fixed plane, and a frozen scaling gate confirms a $1 : \sqrt{p} : p$ certifiability ladder for known, blind, and estimated change in a Gaussian covariance model. The mathematical cores are classical; new here are the optical instantiation, its measured capacity curve, and the engineering doctrine—breakers, noncomposability, jet restoration, wave-optics transfer—of physical realization.

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1 Introduction

A strongly scattering or bandlimited optical channel is usually described by what it destroys. The received wisdom is that a lost coordinate is simply lost. The companion Letter [1] already separates two invariants that this view conflates: reconstruction rank, which can collapse to one, and local change observability, which can stay complete because the first nonzero *jet*—the lowest order at which a scene change perturbs the nuisance-profiled measurement statistics (the “quotient information jet” used below)—fixes the price of observing each direction. That Letter owns the Rank–Jet Separation theorem and the algebraic mean wall of a scrambling channel.

Here we take the complementary and constructive step. Rather than accept an optical sensor’s blindness as a defect, we ask what it *cannot* see, whether that ignorance can be measured, engineered, and certified, and what finite data then permit us to conclude. The mathematical scaffolding is deliberately classical—a symmetry that annihilates the score, the min–max identity for singular values, and the local-asymptotic-normality detection ladder; our object is their optical instantiation and measured certification. Four structural results and a physical realization organize the program:

1. exact measurement symmetries create *walls*—score-null orbits on which no estimator, exposure count, or reconstruction prior can respond (Section 2);
2. calibration measures a wall’s *blindness capacity*—the largest code subspace guaranteed below a declared leak—as a singular spectrum, with a finite metrological envelope and an explicit noncomposability law (Section 3);
3. geometry and projection *select the jet order*, restoring

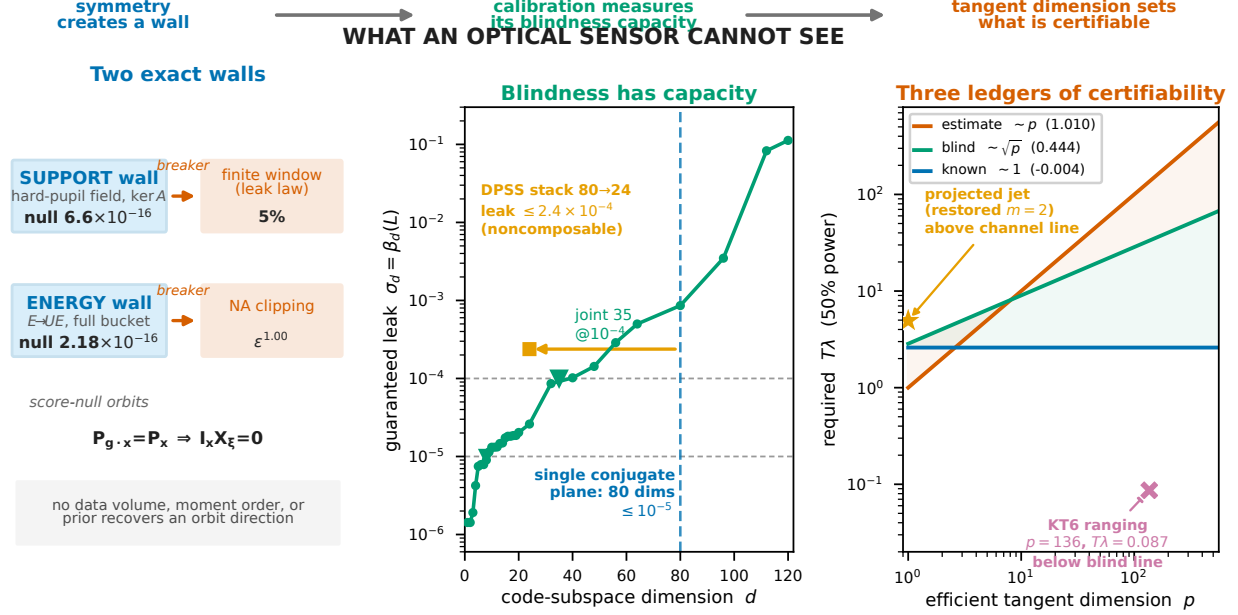
the desired higher-amplitude channel through three physically distinct routes (Section 4);

4. unknown tangent dimension imposes a separate $1 : \sqrt{p} : p$ *certifiability ladder* on known, blind, and estimated change (Section 5);
5. a wave-optics transfer chain carries the algebraic walls to a bench, where a calibrated leak law, a full-Maxwell twin, and apparatus rules fix what physically realizes them (Section 6).

The contribution is not these classical skeletons but their physical realization, in three parts: (a) a *physics mapping* from optical symmetry class to wall type, with two machine-precision instantiations and their specific breakers; (b) a *measured certification apparatus*—an 80-dimensional certified-blind code space, a dimension–leak capacity curve, and a frozen-gate $1 : \sqrt{p} : p$ ladder on a physical instrument model; and (c) an *engineering doctrine* of breakers, noncomposability, and transfer constraints. Each theorem below is billed with the classical result it restates. The sections are organized by theorem, not by chronology; the full verification battery, the honest negatives, and the full proofs are in the Supplement, and every quoted number traces to a single committed artifact.

2 Symmetry walls of an optical measurement

Let a scene x lie on a smooth parameter manifold, let the complete observed record have a dominated law P_x with density $p(y \mid x)$, and let a Lie group G act on the scene with infinitesimal generator X_ξ for a Lie-algebra element ξ .



$\beta_d(L) = \sigma_d(L)$ — and does not automatically compose $1 : \sqrt{p} : p$ — a nonzero signal is not a certifiable signal

Figure 1. What the sensor cannot see. A left-to-right causal spine over three quantitative panels. *Left* — symmetry walls: two exact score-null orbits, the hard-pupil support (field) wall at a maximum leak beyond $2kR$ of 6.6×10^{-16} and the lossless full-aperture energy (unitary) wall at 2.18×10^{-16} (both machine zero), each with its one controlled breaker (finite-window leak law; ϵ^1 NA clipping). *Center* — blindness capacity: the generalized leak singular value σ_d versus code-subspace dimension d (single deterministic joint spectrum), with the single-plane point (80 dimensions $\leq 10^{-5}$ at the fixed conjugate plane), the joint spectrum point (35 dimensions $\leq 10^{-4}$ over four z_1 planes), and the hardware DPSS route ($\leq 2.4 \times 10^{-4}$, whose single-plane SVD capacity does not compose, $80 \rightarrow 24$). *Right* — certifiability ladder: the required $T\lambda \sim 1 : \sqrt{p} : p$ thresholds for known, blind, and estimated change, with the TLSG-measured p -exponents $-0.004/0.444/1.010$; the near-contact KT6 operating point sits below the blind line, and the projected $m = 2$ jet sits above its known-direction channel line. All values are committed frozen artifacts (KT1 @19338a4, IT1 @f13a323, IT4b @5666d1e, IT5 @c8a0487, KT6 @635ff05, TLSG @35d858e).

Theorem 1 (Statistical Noether wall). *If the measurement law is G -invariant, $P_{g \cdot x} = P_x$ for every $g \in G$, then every orbit tangent is an exact score-null direction,*

$$D_x \log p(Y | x)[X_\xi(x)] = 0 \text{ a.s.}, \quad I_x X_\xi(x) = 0, \quad (1)$$

so every f -divergence, Chernoff distance, and quotient information jet between two points of one orbit vanishes: no estimator, moment order, exposure count, or reconstruction prior can distinguish them. Conversely, a smooth constant-rank integrable distribution that is score-null on a connected neighborhood has integral leaves that are local blindness orbits.

Proof sketch. Differentiate $\log p(y | \exp(t\xi) \cdot x) = \log p(y | x)$ at $t = 0$ to get (1); the score Gramian gives $I_x X_\xi = 0$, and equality of the complete laws gives zero divergence at every finite orbit displacement. The converse is Frobenius integrability applied to the score-null distribution. Full statement and the discrete-degeneracy caveat are in Supplement S1. $\square$

The mathematical core is classical: an invariance of a dominated likelihood forces the score to vanish along orbit

tangents and the Fisher information to be singular there [7], and the finite-orbit indistinguishability then follows at once because the laws coincide. What is new is the optical instantiation—which measurement law inherits which scene symmetry—and its two machine-precision realizations. Indeed, the honest content of Theorem 1 is that a symmetry of the wave equation does not by itself create a wall; a wall appears only when the full measurement law inherits the induced parameter action. Two walls of distinct physical type realize it exactly in this idealized band-limited model; on an unfiltered bench each becomes the calibrated $\sim 7 \times 10^{-3}$ residual of Section 6, suppressed by near-conjugate relay imaging.

Support (field) wall. For a bandlimited mean operator A , every $h \in \ker A$ generates a translation $x \mapsto x + h$ that leaves the mean-channel law invariant—the additive group of the unmeasured subspace. A hard Fourier pupil restores this wall to machine precision: the per-variant maximum signed-intensity field leak beyond $2kR$ is 6.6×10^{-16} across all hard-pupil variants and all defocus z_1 , and the ideal periodic-code detector wall is 2.07×10^{-15} for $k > k_p$.

Energy (unitary) wall. A lossless full-aperture bucket measures only the Hilbert-space norm $B(E) = \|E\|_2^2$, so $B(UE) = B(E)$ for every unitary spatial transformation U . A phase-only object change is one orbit of this wall; the measured null is 2.18×10^{-16} at every order and for every (ε, z_2) —the phase-only diagonal subgroup, the only orbit measured here. Within the model assumptions the energy bucket is blind to any unitary spatial-mode transformation; the larger *mode-unitary* wall and its parity, global-phase, and polarization- $SU(2)$ siblings (Supplement S1, Table S1) are corollaries of Theorem 1 that we present as falsifiable predictions of the framework, measured here only for the phase-only subgroup.

The two controlled breakers. Each wall has exactly one designed breaker. NA clipping replaces the identity collector by a projector P_{NA} ; $\|P_{\text{NA}}UE\|^2$ is no longer invariant and the energy wall breaks at first order, with clip-bucket slopes 0.97–1.03 that are monotone in z_2 . A finite object window breaks the support wall: the full twin develops a noncompact pedestal with detector leak beyond $2k_p$ equal to 5.3×10^{-2} (field leak 1.6×10^{-1}), so frequency placement alone gives no wall—an exact detector wall additionally requires a concentration-grade guard (Section 6). These two group actions and their breakers are the first entries of a wall periodic table for an optical chain (Supplement S1); a parity-symmetric chain and a polarization-insensitive bucket supply two further rows.

3 Measuring blindness capacity

An exact wall is a limiting case. In practice one wants the *largest* code space on which a sensor is guaranteed not to respond to a declared nuisance, and a number certifying it. Let a calibrated linear leak operator $L : \mathcal{C} \rightarrow \mathcal{Y}$ map a code to its forbidden mean-channel response, with the code norm whitened by the desired in-band metric, and write its singular values ascending, $\sigma_1 \leq \dots \leq \sigma_n$. Define the best worst-case leak of a d -dimensional code subspace,

$$\beta_d(L) = \min_{\dim S=d} \max_{c \in S, \|c\|=1} \|Lc\|. \quad (2)$$

Theorem 2 (Blindness capacity). $\beta_d(L) = \sigma_d(L)$, attained by the span of the bottom d right singular vectors; no other d -dimensional code family guarantees a smaller worst-case leak. For a generalized relative-leak certificate with desired-response metric $B^\dagger B$, $\beta_d = \sqrt{\lambda_d}$ where $L^\dagger L v = \lambda B^\dagger B v$. If a calibration returns $\hat{L}$ with $\|L - \hat{L}\|_{\text{op}} \leq \delta$, the bottom- d subspace of $\hat{L}$ obeys the metrological envelope

$$\max_{c \in S_d, \|c\|=1} \|Lc\| \leq \sigma_d(\hat{L}) + \delta. \quad (3)$$

Proof sketch. Equation (2) is the Courant–Fischer min–max characterization of σ_d : the bottom- d subspace attains σ_d ,

and any d -dimensional subspace meets the orthogonal complement of the bottom $d - 1$ singular vectors, so it contains a unit vector with leak at least σ_d . The generalized form is the same statement for the pencil $(L^\dagger L, B^\dagger B)$; (3) is Weyl’s inequality. Supplement S2. $\square$

The identity $\beta_d = \sigma_d$ is the Courant–Fischer min–max characterization of singular values [8] and coincides with the null-space resolution analysis of Backus–Gilbert and truncated SVD [9]; what is new is the *measured* dimension–leak spectrum of a specific optical leak operator and its calibration-transfer certificate. Theorem 2 makes “cannot see” a measured spectrum. At the fixed relay-conjugate plane, 80 code dimensions are certified below 10^{-5} (and below 10^{-4})—a single-plane blindness-capacity point, not an incidental code-design success. Enforcing invisibility *jointly* over four defocus planes $z_1 \in \{0, 2, 5, 10\}$ mm costs capacity: 35 dimensions remain below 10^{-4} jointly (8 below 10^{-5}), so the single-plane requirement buys 45 extra dimensions. The full ascending spectrum runs from $\sigma_1 = 1.43 \times 10^{-6}$ through $\sigma_{64} = 4.99 \times 10^{-4}$ to $\sigma_{120} = 1.13 \times 10^{-1}$; the plotted capacity curve of Fig. 1. In the $d = 64$ subspace the worst-direction leak is 4.99×10^{-4} , while a random in-subspace code leaks 1.68×10^{-4} in RMS—threefold below the worst direction and ridge-stable (d at 10^{-4} is 35 across regularizers $0/10^{-8}/10^{-6}$). Certified-blind codes retain a second sensing channel: the random-to-blind eigenvalue ratio is only 1.20 ($\ll 3$), yet the operational mean leak is $219 \times$ lower (2.58×10^{-4} versus 5.64×10^{-2}).

The envelope (3) is what upgrades a numerical null space to a certified instrument property (Fig. 3, right). On a fresh calibration/test split the simultaneous $\sigma_{64}(\hat{L}) + \delta$ envelope has coverage 1.000 (24/24 fresh draws; $\sigma_{64} = 3.4 \times 10^{-8}$, $\delta = 1.07 \times 10^{-2}$, worst fresh bottom-64 leak 1.99×10^{-6} , six orders below the envelope), and at least 64 conjugate-plane dimensions stay certified below 10^{-4} on fresh physical-law draws, with a measured minimum of 80. This fresh-draw minimum (at 10^{-4}) coincides numerically with, but is a distinct and stronger statement than, the single-plane deterministic capacity (80 dimensions at 10^{-5}) reported above.

4 Restoring the desired jet

The exact walls of Section 2 are the endpoints of a continuous mechanism. Let the profiled Chernoff distance carry two leading operators, $C(\varepsilon, \gamma) = c_1 \gamma^{2a_1} \varepsilon^{2m_1} + c_2 \gamma^{2a_2} \varepsilon^{2m_2}$ with $0 < m_1 < m_2$ and a geometric gain γ (the contact distance z_2 in the near-contact case), and set $y = (c_2/c_1) \gamma^{2(a_2-a_1)} \varepsilon^{2(m_2-m_1)}$. The effective amplitude order $m_{\text{eff}} = \frac{1}{2} d \log C / d \log \varepsilon = (m_1 + m_2 y) / (1 + y)$ obeys the exact flow

$$\frac{dm_{\text{eff}}}{d \log \varepsilon} = 2(m_{\text{eff}} - m_1)(m_2 - m_{\text{eff}}), \quad (4)$$

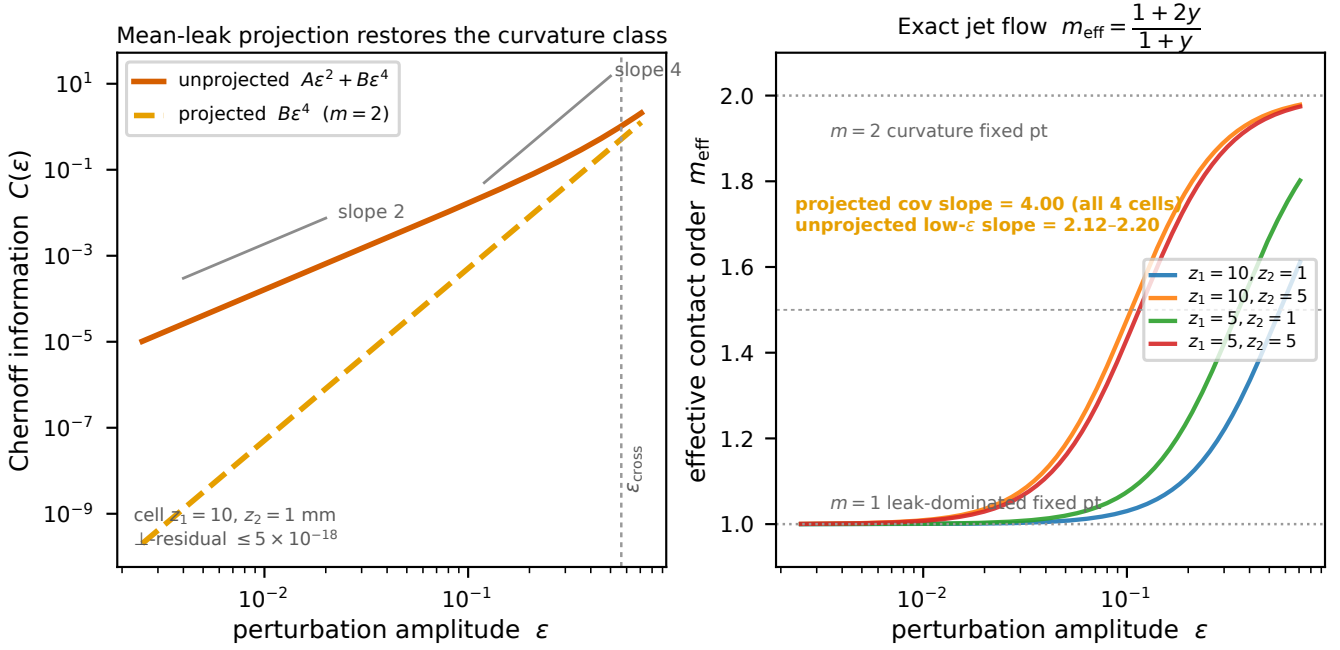


Figure 2. Jet transmutation. *Left:* the profiled covariance distance $C(\varepsilon) = c_1 \varepsilon^2 + c_2 \varepsilon^4$, an analytic two-coefficient reconstruction from the committed fitted coefficients of the four KT2 cells (not a raw $C(\varepsilon)$ sweep). Unprojected, a first-order mean leak ($c_1 > 0$) gives slope 2 ($m = 1$) at small amplitude; projecting the measured mean-leak tangent zeroes c_1 and leaves slope 4 ($m = 2$) everywhere, with the crossover at $\varepsilon_{\text{cross}} = \sqrt{c_1/c_2}$. The projected slope 4 is the exact analytic consequence of zeroing c_1 —a consistency check, not an independent measurement—and the four cells reproduce it to 4.00. *Right:* the exact logistic jet flow $dm_{\text{eff}}/d \log \varepsilon = 2(m_{\text{eff}} - 1)(2 - m_{\text{eff}})$ between the integer fixed points 1 and 2; the unprojected low-amplitude slopes are 2.12–2.20. Cross-anchor: the companion integer contact orders 2.038/4.000 (Ref. [COMPANION]). Source: KT2 @05f1029.

with fixed points m_1 and m_2 .

Corollary 1 (Jet transmutation). *A lower-order symmetry-breaking channel ($m_1 = 1$) drives the strict local observability class to $m_{\text{eff}} \rightarrow 1$ as $\varepsilon \rightarrow 0$; setting its coefficient to zero by an exact wall, a certified code, or a nuisance projection places the experiment on the higher-order fixed manifold $m_2 = 2$ without changing the scene perturbation. In the near-contact optical case $m_1 = 1, a_1 = 0, m_2 = 2, a_2 = 2$, so the crossover invariant is $\varepsilon z_2^2 = \text{const}$, i.e. $z_{2,*} \propto \varepsilon^{-1/2}$.*

Three physically distinct routes realize the corollary, and they are alternatives, not factors.

Route 1 — optical projection. Projecting out the measured mean-leak tangent zeroes the first-order coefficient c_1 , so the covariance channel becomes a pure ε^4 monomial whose log–log slope is 4 by construction; this consistency check (Fig. 2) returns 4.00 in all four cells, against an unprojected leak-dominated slope of 2.12–2.20, with orthogonality residual $\sim 10^{-18}$. A generic direction, by contrast, stays sub-quartic (slopes 1.58–2.28).

Route 2 — code-space projection. Projecting the covariance score off the leak/law span drives the nuisance overlaps to zero (amplitude $0.36 \rightarrow 0$, medium $0.20 \rightarrow 0$) while retaining 86–99% of the target information.

Route 3 — hardware concentration. A discrete prolate spheroidal (DPSS) [15] render-and-object window holds the beyond- $2k_p$ witness leak below 2.4×10^{-4} uniformly in z_1 (maximum 2.38×10^{-4}), $349\times$ below the no-window value 8.3×10^{-2} ; a naive separable Tukey window reaches only $\sim 4 \times 10^{-2}$ (rim-limited).

These routes do not stack. For several leak routes $L_1, \dots, L_k$ the simultaneous root-sum-square certificate is governed by the stacked Gram operator, so

$$\beta_d^2 = \lambda_d^\dagger \left(\sum_i w_i^2 L_i^\dagger L_i \right), \quad (5)$$

which is not multiplicative in the individual certificates. Concretely, the DPSS hardware route does not compose with the SVD code route: the single-plane certified dimension drops from 80 to 24 at 10^{-4} , because the window rotates and reshapes the operator and trades code-space capacity. A second, independent witness of the same law is the Tukey apodization, whose joint certified dimension drops from 35 to 24 and which is $93\times$ worse at $d = 64$ (apodized relative leak 4.66×10^{-2} versus 4.99×10^{-4} ; a rim effect). Independent hardware modifications change the leak operator and its metric and must be recalibrated jointly.

5 What finite data can certify

A wall says a change is invisible; blindness capacity says how large a subspace is guaranteed invisible; the jet order says how a visible change scales. A fourth, independent price is set by finite data and unknown direction. Whiten the null covariance, profile every admitted nuisance tangent, and let the efficient covariance displacement be $A_\perp$ in a symmetric tangent space of dimension p , with efficient signal $\lambda = \frac{1}{2}\|A_\perp\|_F^2$. For T independent Gaussian banks $Y_t \sim \mathcal{N}(0, \Sigma_\varepsilon)$ the aggregate score experiment is locally $Z_T \approx \mathcal{N}(\sqrt{T}\theta, I_p)$ with $\|\theta\|^2 = \lambda$.

Theorem 3 (Three-ledger certifiability). *For a Gaussian covariance model reduced by local asymptotic normality, under this nuisance-profiled reduction the per-task bank budget for fixed nontrivial power obeys*

$$T\lambda \sim \begin{cases} 1, & \text{known direction (matched score),} \\ \sqrt{p}, & \text{blind existence (unknown direction),} \\ p, & \text{estimate or localize the direction,} \end{cases} \quad (6)$$

a $1 : \sqrt{p} : p$ ladder. Composed with a quotient jet $A_\perp = \kappa \gamma^a \varepsilon^m B$, the known-direction budget is $T_{\text{known}} \asymp \kappa^{-2} \gamma^{-2a} \varepsilon^{-2m}$, and the blind and estimation budgets multiply it by $\sqrt{p}$ and p ; adaptation changes the prefactor, never the physical exponent $2m$.

Proof sketch. The matched score $u^\top Z_T$ separates the hypotheses at $T\lambda = O(1)$. For the blind ledger the invariant $\|Z_T\|^2$ is χ_p^2 versus $\chi_p^2(T\lambda)$, separated when $T\lambda \gg \sqrt{p}$; a uniform spherical mixture gives a matching minimax lower bound, since its χ^2 -divergence from the null scales as $T^2 \lambda^2 / p$ and vanishes below $T\lambda \sim \sqrt{p}$. The efficient estimator $\hat{\theta} = Z_T / \sqrt{T}$ has risk p/T , matching the Cramér–Rao bound, so relative error is controlled only at $T\lambda \sim p$. Supplement S4. $\square$

The three scalings are classical—matched-filter detection at $O(1)$, the Ingster–Suslina high-dimensional detection boundary at $\sqrt{p}$ [10, 14], and the Cramér–Rao bound at p [7]; what is new is their realization as a single measured operating map on an optical instrument model.

A dedicated scaling gate (TLSG) confirms the ladder without a repair round. The fitted p -exponents for the known, blind, and estimation tasks are $-0.004/0.444/1.010$ against the frozen targets $0, \frac{1}{2}, 1$. The matched-score 50%-power contours collapse against $T\lambda$ with coefficient of variation $CV = 0.125$; the blind contours collapse against $T\lambda/\sqrt{p}$ ($CV = 0.124$) and *fail* against $T\lambda$ alone ($CV = 0.628$), the signature of the $\sqrt{p}$ adaptation penalty; the estimation risk collapses against $T\lambda/p$ ($CV = 0.035$, mean risk $\cdot T\lambda/p = 0.985$, i.e. the Cramér–Rao floor p/T).

Each exponent is a single log–log fit over $n = 4$ distinct tangent dimensions $p \in \{10, 36, 136, 528\}$ (8 cells, frozen seed 20260724), with in-sample $R^2 = 0.003, 0.983,$

0.9995 for the known, blind, and estimation tasks (deterministic recomputation from the committed points). A post-gate 10-seed bootstrap (unmodified runner, post-hoc robustness) confirms the known and estimation exponents—mean -0.011 , 95% CI $[-0.044, 0.012]$, and mean 0.996 , CI $[0.985, 1.008]$, each bracketing its target—while the blind exponent, mean 0.454 , CI $[0.446, 0.465]$, has an interval that *excludes* the asymptotic $\frac{1}{2}$: a systematic finite- p deviation of $\approx 9\%$ over this fit range, not sampling noise. The frozen gate verdict is unaffected—all 10 seeds pass every bar inside the ± 0.15 band—and the $0 : \frac{1}{2} : 1$ ladder-order separation is unambiguous.

The ladder also predicts a negative for the right reason. The near-contact channel-ratio operating point (KT6) sits below the optimal blind tier: its blind detection power is 0.051 at a false-positive rate of 0.05 (rising to 0.056 and 0.134 at $10\times$ and $100\times$ exposure), with $p = 136$ and $T\lambda = 0.0866$. The oracle law is nonetheless intact—the near-contact ratio $J_{\text{cov}}/J_{\text{mean}} \propto z_2^4$ with fitted exponents 3.99 – 4.08 —but the blind signal-to-floor ratio is $5.4 \times 10^{-3} \ll 1$, below the Wishart M^2/T floor [16], so the estimator is killed. This is a boundary of the third ledger, not a failure of apparatus: the projected $m = 2$ covariance channel of Section 4 (slope 4.00) is by contrast a *known*-direction observable, and sits above its channel line in Fig. 1.

6 Wave-optics transfer and apparatus constraints

The theorems above are algebraic; a bench is not. On the unfiltered complementary-DMD twin the exact mean wall becomes a reproducible calibrated residual: a leak law with a $\sim 7 \times 10^{-3}$ conjugate-plane floor and a weakly super-linear defocus term, established and owned by the companion Letter [1] (End Matter C), which we take here as the boundary condition for the transfer results new to this paper. That floor is an instrument constant set by fill-factor and macro-pixel staircase effects (moving 6% and 50% , with $L_{\text{pix}} = 7.15 \times 10^{-3}$) that the pupil does not reduce; its four independent estimates—leak-law intercept 6.94×10^{-3} , pixelation constant 7.15×10^{-3} , deterministic conjugate-plane leak 6.8×10^{-3} , and analog-PWM baseline 7.0×10^{-3} —are consistent readings of one $\sim 7 \times 10^{-3}$ floor from different tests, not scatter on one measurement. Deterministic beyond-band leak grows with defocus, from 6.8×10^{-3} at the conjugate plane to 5.4×10^{-2} at $z_1 = 10$ mm and 8.9×10^{-2} at $z_1 = 20$ mm (Fig. 3, left); near-conjugate relay imaging suppresses the propagation term. A full-Maxwell check *scopes* the reduced model rather than validating it wholesale: over $n = 8$ realizations the thin-phase-screen speckle contrast 0.467 agrees with a two-dimensional finite-element twin (COMSOL 6.3 Wave Optics, 2D TE) contrast 0.395 to 15.5% in the ensemble mean, but the speckle grain size disagrees by 65% (thin-screen $3.88 \mu\text{m}$ versus FEM

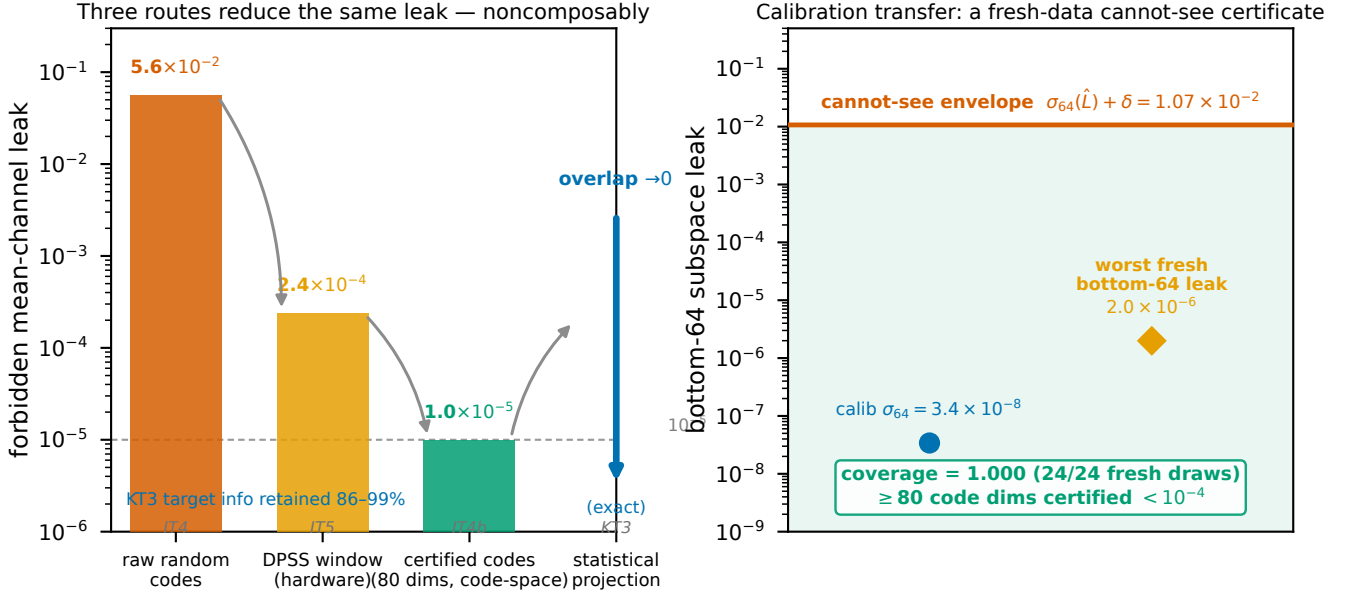


Figure 3. Restoration routes and calibration transfer. *Left:* the descending leak ladder of the three physically distinct routes plus the exact statistical projection, all reducing the forbidden mean-channel response—raw random codes 5.6×10^{-2} , DPSS window 2.4×10^{-4} , certified codes $\leq 10^{-5}$ (single-plane, 80 dimensions), statistical projection to 0 (exact)—which are alternatives, not factors. *Right:* calibration transfer (Eq. (3)) on a fresh calibration/test split: the simultaneous cannot-see envelope is 1.07×10^{-2} , the worst fresh bottom-64 leak is 2.0×10^{-6} (six orders below), coverage is 100% on 24 fresh draws, and at least 80 code dimensions remain certified below 10^{-4} at the conjugate plane. Sources: IT4 @33b8ef2, IT5 @c8a0487, IT4b @5666d1e, KT3 @2430166, TLSG @35d858e.

1.35 μm , a $2.9\times$ overestimate), consistent with the multiple-scattering decorrelation the thin screen omits. The contrast statistics therefore transfer; the grain-scale statistics do not, and we make no grain-scale claim from the reduced model. The covariance signal grows as $\lambda \propto M^{1.8-1.9}$ in code count (fitted exponent 1.78 at mid range, 1.89 near contact), and the geometry-matched developed cell extrapolates to ≈ 768 code banks at 2% error for $M = 128$, inside the sealed band [513, 2996].

Two apparatus rules follow. First, because a separable Tukey guard reduces the detector tail only $\sim 8\text{--}40\times$ and never to 10^{-12} , an exact physical detector wall requires a concentration-grade (DPSS) guard together with the hard Fourier pupil—the choice of Section 4, Route 3. Second, binary rendering must use full-period pulse-width modulation with whole-period bucket integration: Bayer-ordered spatial dithering doubles the conjugate-plane leak, from 7.0×10^{-3} to 1.38×10^{-2} , above the 10^{-2} tolerance.

7 Discussion

Blindness with a symmetry origin is not a defect to be hidden but a quantity to be calibrated, engineered, and certified. The four objects are independent: a symmetry decides whether a change exists in the record, the jet order decides how its signal scales, the singular spectrum decides how large a code space is guaranteed blind, and the tangent dimension decides whether that signal can be blindly certified. A physical sensor can therefore *publish* a cali-

brated dimension–leak frontier describing the largest code space on which it is guaranteed not to respond to a declared nuisance, backed by the metrological envelope (3) rather than an uncertified numerical null. The closest structural cousins—decoherence-free subspaces [3, 4], dynamical decoupling [5], and unobservable subspaces in nonlinear control [6]—share the kernel idea; the optical novelty is the hardware-calibrated, task-selective blindness with a measured capacity curve and a finite-sample certifiability ladder. The program is optics-first by design: without a bench instrument or a second non-optical physical realization the synthesis belongs in Optica, and the memory-effect phase diagram [2] (which requires a genuine thickness or stacking axis) is deferred. The same symmetry-first accounting extends to quantum-limited sensing, where a measurement symmetry likewise sets what no estimator can resolve [17]. The honest negatives that bound the scope—a segmented bucket that breaks the mean wall, and a coincidence veto that repairs medium false alarms while retaining only 24% of scene power—appear with the positives in Supplement S5.

Relation to prior work. We claim none of the classical cores. The capacity identity $\beta_d = \sigma_d$ is Courant–Fischer min–max [8] and coincides with the null-space resolution analysis of Backus–Gilbert and truncated SVD [9]; the Noether wall is the invariance $\Rightarrow$ singular-Fisher fact [7]; and the $1 : \sqrt{p} : p$ ladder combines matched-filter detection, the Ingster high-dimensional detection boundary [10], and the Cramér–Rao bound, its decision-theoretic “cannot see” be-

ing Barrett–Myers ideal-observer detectability [11]. Declaring a guaranteed-invisible code subspace with a bounded leak is the wiretap / physical-layer-security framing [12], and the certifiability language is adjacent to compressed-sensing null-space and restricted-isometry certificates [13]. The delta this paper adds to each is the same: a hardware-calibrated, task-selective capacity curve measured on an optical instrument, not a new inequality.

Back matter

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Disclosures. The authors declare no conflicts of interest.

Data availability. The committed frozen artifacts underlying every numerical claim, the claim–source matrix, and the figure scripts are available at [URL]; the full-Maxwell twin additionally requires a COMSOL 6.3 (Wave Optics) license to reproduce. See Supplement 1 for the full verification battery, proofs, and the TLSG gate table.

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